

Validated Lie Group Variational Integrator-Based Simulator for Passive Magnetic Attitude Control of the GASRATS CubeSat

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ABSTRACT

Passive magnetic attitude control (PMAC) systems are selected for integration in CubeSat designs on account of their simplicity and lack of power consumption. These systems function by simultaneously dissipating a satellite's rotational kinetic energy as heat and orienting it according to the Earth's magnetic field. This process is critical for establishing steady-state attitude, a stable rotational state required for conducting mission operations and enabling downlink to ground stations. Comprehensive attitude simulations are necessary for confirming the reliability of a PMAC system to provide timely attitude convergence for a wide range of deployment velocities and orientations. An attitude damping phase precedes steady-state, and the duration may last weeks or longer. As simulation timespans of at least one month are required to prove a stable convergence, standard adaptive Runge-Kutta (4, 5) solvers (RK45) are less suitable because of the accumulation of numerical error. Lie group variational integrators (LGVI) offer a compelling alternative. LGVIs demonstrate excellent geometric conservation properties and maintain attitude matrix orthogonality, resulting in improved physical accuracy. A three-step validation chain is followed to verify a developed LGVI-based attitude simulator, comprising validations against literature, a cross-framework verification with an RK45-based simulator, and an invariant-preservation analysis. In a torque-free scenario, the LGVI is found to maintain a relative angular momentum error on the order of 10^{-13} while halving the execution time of an RK45 integrator configured with a comparable timestep. The validated LGVI simulator is employed to model the performance of eight PMAC configurations for the Get Away Special Radio and Antenna Transparency Satellite in a 1,001-trial Monte Carlo analysis. The results inform the final design recommendation of a four-rod, 1.5U configuration and indicate the unexpectedly poor damping performance of 1U configurations. It is determined that this phenomenon is associated with inertial symmetry that limits the transfer of rotational kinetic energy between body axes.

INTRODUCTION

Lie group variational integrators (LGVIs) are selected for implementation in the next generation of attitude simulations for the Get Away Special Radio and Antenna Transparency Satellite (GASRATS) by virtue of their reported physical accuracy and computational efficiency in comparison with Runge-Kutta (RK) integrators. LGVIs are of interest to attitude simulations of passive magnetic attitude control (PMAC) systems because they provide favorable energy behavior and preserve the orthogonal structure of the attitude matrices. The LGVI is a modern integration method that was pioneered in the 1990s and advanced by Lee et al.

(2005) for attitude dynamics with a dependent potential.¹ LGVIs have been used in PMAC simulations of past CubeSat missions, such as the University of Michigan's first Radio Aurora Explorer (RAX), on account of their energy preservation over long simulation timespans. As PMAC simulations for CubeSats operating in low Earth orbit (LEO) target durations of one month or longer to prove rotational stability, RK integrators are not ideal due to the accumulation of numerical error. Although LGVIs offer improvements in energy behavior and efficiency over standard integration methods, their complexity renders the production of valid implementations a challenging endeavor. The purpose of the LGVI theory presented in this paper is

to provide an implementation-oriented bridge to the source literature, thereby supporting the adoption of LGVIs by other CubeSat teams. Following this, a developed LGVI-based attitude simulator is subject to a three-part validation and is employed in a Monte Carlo analysis, assessing the damping performance of eight PMAC system configurations.

A LIE GROUP VARIATIONAL INTEGRATOR

Geometric and Numerical Foundations

The desirable energy- and structure-preserving qualities of the LGVI stem from its underlying architecture. Contrasting LGVIs with standard RK integrators highlights the nuances of the variational integration method. For standard adaptive RK methods—such as the Runge-Kutta (4, 5) solver (RK45)—used for attitude modeling in a quaternion framework, the state trajectory is approximated by computing fourth- and fifth-order accurate estimates of the next state at each timestep. Instead of approximating the state trajectory, the LGVI approximates the system’s action, and discrete Lagrangian equations of motion are derived from a discrete action sum via Hamilton’s principle. A Legendre transformation yields the Hamiltonian form, providing an implicit momentum equation that is solved for the rotation increment needed for the kinematic update. The attitude evolves natively on the manifold of valid rotation matrices, the special orthogonal group $SO(3)$. Thus, the matrices used for attitude representation remain orthogonal. Derived from a variational principle, the LGVI provides symplectic update mapping for conservative potentials and exhibits desirable long-term energy behavior for systems with dissipative magnetic hysteresis.

One downside of RK45 attitude simulations that employ quaternion kinematics is norm drift, meaning that the magnitude of the unit quaternion shifts off unit during integration. Error-introducing normalization steps are required to force quaternions back to unit length, impacting the accuracy of the simulation. RK45 integrators typically advance a quaternion as four coordinates in 4D vector space. In the integration step, the derivative at a point is calculated, and a tangential move is made that

deviates off the unit quaternion hypersphere. A normalization step reprojects the non-unit quaternion back onto the unit sphere, resulting in numerical drift that affects simulation accuracy. Fig. 1 provides a geometric illustration of normalization for three integration schemes.

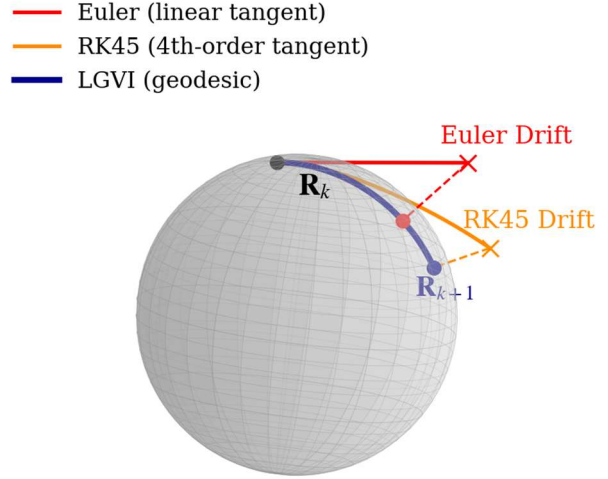


Figure 1. Position Update Mapping for Three Integration Methods Represented on the S^3 Manifold

In contrast, the LGVI method evolves a rotation matrix of $SO(3)$ strictly on the $SO(3)$ manifold by a multiplicative update constructed via exponential mapping. $SO(3)$ is the group of all valid rotation matrices that satisfy the conditions $R^T R = I$ and $\det(R) = +1$. The integrator utilizes exponential mapping to generate rotation updates on the manifold, producing geodesic paths between timesteps. LGVIs for attitude modeling employ rotation matrices instead of quaternions due to the numerical issues that arise from quaternion sign duplicity, meaning that the negative and positive alternates of a quaternion represent the same physical orientation. This produces complications in the integration methods that motivate the selection of rotation matrices for attitude representation in LGVIs.

It may seem counterintuitive that LGVIs are notably computationally efficient in comparison to RK45 integrators—LGVIs use a specialized Newton iteration to solve an implicit momentum balance equation as well as heavier 3×3 rotation matrices. Conversely, researchers from the University of Michigan

demonstrated that the LGVI approach offers improvements on simulation runtime in comparison to RK45. Work conducted by Lee et al. (2011) on simulations for the RAX CubeSat reported that their LGVI integration speed was 3.8 times faster than an RK45 solver configured with corresponding timesteps for long duration simulations, excluding hysteresis torques.² For an RK45 integrator to achieve similar energy preservation to LGVIs, its timestep would have to decrease. This would increase runtime and further widen the efficiency gap between the two integrators. Thus, an LGVI is notably efficient in comparison to a RK45 configured to provide similar physical fidelity. The efficiency of the LGVI is advantageous for the computationally intensive Monte Carlo analysis later discussed that tests many hundreds of initial conditions.

Mathematical Formulations

The theory of the LGVI is presented in a way that directly ties to its MATLAB implementation and is based on the derivations presented by Lee et al. (2005). The purpose of this explanation is to serve as a bridge to the LGVI source literature by providing an intuitive guide for undergraduates. This section highlights important steps in the derivation of the discrete Euler-Lagrange equation and the implicit momentum balance. The residual of the implicit momentum balance is provided, which forms the core of the Newton iteration. Therein, a rotation increment required for the attitude update is found that satisfies the residual equation. A systematic understanding of Hamilton's principle is essential for implementing LGVIs in MATLAB simulations and carries significant value due to its widespread occurrence throughout physics, including the action-based derivation of the Einstein field equations. By enforcing that the system follows a path such that its discrete action is stationary, the resulting integrator generates trajectories that are consistent with the physical laws governing the system's natural motion.

The derivation of the discrete Euler-Lagrange equation begins with the attitude kinematic equation. The configuration space is $SO(3)$, and the kinematic equation is given by

$$\dot{R} = RS(\Omega), \quad (1)$$

where $R \in SO(3)$ is the attitude matrix and $S(\Omega)$ is the skew-symmetric matrix associated with the angular velocity $\Omega \in \mathbb{R}^3$ in the body frame. Attitude simulations require discrete versions of such equations. Using the approximation $S(\Omega_k) = (F_k - I_{3 \times 3})/h$, the discrete kinematic equation can be expressed as

$$R_{k+1} = R_k F_k, \quad (2)$$

with the rotation increment matrix $F_k = R_k^T R_k$. LGVIs follow an energy-based formulation and are governed by discrete equations of motion in Lagrangian form. A Lagrangian can be defined as the difference between kinetic energy T and potential energy U , and a general form for the configuration space $SO(3)$ is given by

$$L(R, \Omega) = T(\Omega) - U(R). \quad (3)$$

For a rigid body, kinetic energy is the sum of the motion of every particle within that body, and the continuous Lagrangian equation can be expressed as

$$L(R, \Omega) = \frac{1}{2} \int_{Body} \|\Omega \times \tilde{\rho}\|^2 dm - U(R), \quad (4)$$

where $\tilde{\rho} \in \mathbb{R}^3$ represents a vector from the center of mass of the satellite to an individual mass element dm , and the velocity of a mass element is $\Omega \times \tilde{\rho}$.

This continuous equation is altered from vector to matrix form to provide compatibility with Lie group operations. Skew mapping replaces the cross product of Ω and $\tilde{\rho}$, and the trace identity $\|x\|^2 = \text{tr}(xx^T)$ is applied. The non-standard moment of inertia matrix J_d is preferred over the standard moment of inertia J for matrix operations.

$$J_d = \int_{Body} \tilde{\rho} \tilde{\rho}^T dm \in \mathbb{R}^{3 \times 3}$$

is substituted, resulting in the continuous Lagrangian for a system evolving on $SO(3)$, which can be expressed as

$$L(R, \Omega) = \frac{1}{2} \text{tr}[S(\Omega) J_d S(\Omega)^T] - U(R). \quad (5)$$

The discrete Lagrangian is found by approximating the action integral over one timestep with a trapezoidal rule to maintain second-order numerical accuracy. The discrete approximation of the skew-symmetric angular velocity is substituted, where h is the duration of one timestep:

$$L_d(R_k, F_k) \approx \int_{t_k}^{t_{k+1}} \frac{1}{2} \text{tr}[S(\Omega) J_d S(\Omega)^T] - U(R) dt,$$

$$L_d(R_k, F_k) = \frac{1}{h} \text{tr}[(I_{3 \times 3} - F_k) J_d]$$

$$- \frac{h}{2} U(R_k) - \frac{h}{2} U(R_{k+1}). \quad (6)$$

Hamilton's principle states that the action is stationary along the physical trajectory of the system, such that its first variation is zero for a given time interval. The action is defined by the integral of (5) from initial time t_0 to final time t_f and is given by

$$\mathfrak{S} = \int_{t_0}^{t_f} \frac{1}{2} \text{tr}[S(\Omega) J_d S(\Omega)^T] - U(R) dt. \quad (7)$$

The discrete action sum is the discrete counterpart of (7). It can be expressed as

$$\mathfrak{S}_d = \sum_{k=0}^{N-1} \frac{1}{h} \text{tr}[(I_{3 \times 3} - F_k) J_d]$$

$$- \frac{h}{2} [U(R_k) + U(R_{k+1})]. \quad (8)$$

The varied discrete action sum is found by substituting the varied rotation increment $F_k^\epsilon = e^{-\epsilon \eta_k} F_k e^{\epsilon \eta_{k+1}}$, where $\epsilon \in \mathbb{R}$ and the variation $\eta \in \mathfrak{so}(3)$. The result can be expressed as

$$\mathfrak{S}_d^\epsilon = \sum_{k=0}^{N-1} \frac{1}{h} \text{tr}[(I_{3 \times 3} - e^{-\epsilon \eta_k} F_k e^{\epsilon \eta_{k+1}}) J_d]$$

$$- \sum_{k=0}^{N-1} \left\{ \frac{h}{2} U(R_k e^{\epsilon \eta_k}) + \frac{h}{2} U(R_{k+1} e^{\epsilon \eta_{k+1}}) \right\}. \quad (9)$$

The first variation of discrete action is found by differentiating (9) and evaluating the perturbation parameter ϵ at zero. The boundary condition $\eta_0 = \eta_N = 0$ is applied, and the shorthand notation $U_k = U(R_k)$ for discrete potential energy is adopted. According to Hamilton's principle the first variation is set equal to zero, yielding

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathfrak{S}_d^\epsilon = \sum_{k=1}^{N-1} \text{tr} \left[\eta_k \left\{ \frac{1}{h} (F_k J_d - J_d F_{k-1}) \right. \right.$$

$$\left. \left. + h R_k^T \frac{\partial U_k}{\partial R_k} \right\} \right] = 0. \quad (10)$$

Because the discrete variation η_k is a skew-symmetric matrix, the expression within the braces of (10) is symmetric. Thus, matrix identity $A - A^T = 0$ can be applied, yielding the Euler-Lagrange equation:

$$\frac{1}{h} (F_{k+1} J_d - J_d F_k - J_d F_{k+1}^T + F_k^T J_d)$$

$$= h \left(\frac{\partial U_{k+1}}{\partial R_{k+1}}^T R_{k+1} - R_{k+1}^T \frac{\partial U_{k+1}}{\partial R_{k+1}} \right). \quad (11)$$

(11) paired with (2) forms the discrete equations of motion in Lagrangian form. The left-hand side of (11) is the discrete kinetic variation term, representing the evolution of the system's angular momentum. The right-hand side of (11) represents the discrete torque generated by the internal potential U .

Applying a discrete Legendre transformation introduces Π_k to represent discrete angular momentum. This transition to a Hamiltonian system is made by setting the skew form of discrete momentum equal to the directional derivative of the discrete Lagrangian. This results in the following equation in which $M_k \in \mathbb{R}^3$ represents the discrete torque acting on the body in Hamiltonian form:

$$S(\Pi_k) = \frac{1}{h} (F_k J_d - J_d F_k^T) - \frac{h}{2} S(M_k). \quad (12)$$

Lee et al. (2005) derive M_k from an internal gravitational potential; here, M_k is evaluated from external magnetic torques. By performing algebraic manipulations and scaling by timestep h , a transition to the implicit momentum balance is made:

$$h S \left(\Pi_k + \frac{h}{2} M_k \right) = F_k J_d - J_d F_k^T. \quad (13)$$

Whereas the residual of (13) is used to find the rotation increment F_k required for the attitude update, a discrete momentum update is needed to compute the angular momentum for the next timestep:

$$\Pi_{k+1} = F_k^T \Pi_k + \frac{h}{2} F_k^T M_k + \frac{h}{2} M_{k+1}. \quad (14)$$

(14), (13), and (2) form the discrete equations of motion in Hamiltonian form. The Hamiltonian formulation differs from its Lagrangian counterpart by describing a system in terms of position and momentum versus position and velocity. A vectorization operation is performed on (13) to flatten its 3×3 matrices into 3×1 column vectors. This operation translates all non-standard inertia matrices J_d back to standard J . F_k is replaced by the rotation increment vector $f_k \in \mathbb{R}^3$. F_k and f_k are related by Rodrigues' formula, which can be expressed using $c_1 = \sin \|f_k\| / \|f_k\|$ and $c_2 = (1 - \cos \|f_k\|) / \|f_k\|^2$:

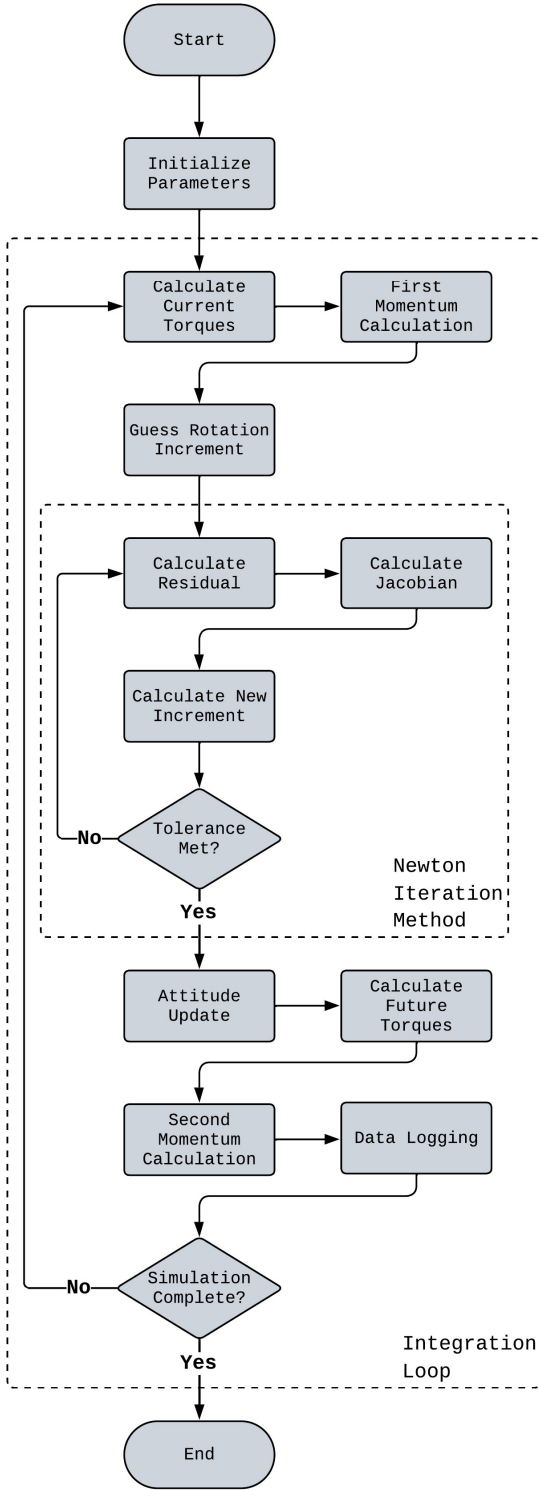


Figure 2. Architecture of an LGVI-Based Attitude Simulator

$$F_k = e^{S(f_k)} = I_{3 \times 3} + c_1 S(f_k) + c_2 S(f_k)^2. \quad (15)$$

Finally, the resulting equation can be expressed in terms of the residual G :

$$G = (c_1 J f_k + c_2 f_k \times J f_k) - \left(h \Pi_k + \frac{h^2}{2} M_k \right). \quad (16)$$

The residual equation forms the core of the Newton iteration. It is used to obtain the rotation increment vector f_k required for the attitude update. For (16), Rodrigues' formula replaces the exponential function $F_k = e^{S(f_k)}$ with a trigonometric equation that can be quickly calculated by a computer. As F_k is formulated via exponential mapping, R_k remains an orthogonal matrix. This provides the structure-preserving property of the LGVI, ensuring the attitude matrix evolves strictly on the $SO(3)$ manifold. By applying a discrete variational principle—enforcing the stationary condition $\delta \mathfrak{S}_d = 0$ of Hamilton's principle—the integrator is provided with symplectic attitude and angular momentum update mapping for conservative potentials, meaning that the geometric relationship between position and momentum is preserved. These methods supply the LGVI's desirable long-term energy behavior that is advantageous for extended PMAC simulations.

Computational Architecture

A practical implementation of the LGVI theory in a MATLAB-based algorithm follows a leapfrog integration method in its kick-drift-kick form. The momentum update is implemented via two half-step calculations that frame the attitude kinematic update. The architecture of this algorithm is illustrated in Fig. 2.

First, initial conditions of the simulation are set before the main integration loop commences. This includes initializing the timestep, inertial matrices, hysteresis rod parameters, and attitude matrices and momentum vectors. Second, within the main integration loop, simulation time is passed into a circular orbital propagator function that returns the satellite position r_{ECI} in the Earth-centered inertial (ECI) frame. This position is converted to the Earth-centered Earth-fixed (ECEF) frame position r_{ECEF} via the transformation matrix R_{EI} by $r_{\text{ECEF}} = R_{\text{EI}}(t)r_{\text{ECI}}$. The latitude, longitude, and altitude values from r_{ECEF} are input to the

International Geomagnetic Reference Field (IGRF) model, returning the magnetic flux density vector in local North-East-Down coordinates, B_{NED} [nT]. This flux density is rotated to ECEF and converted to Tesla, yielding B_{ECEF} . A conversion to field strength H_{ECEF} is made via $H_{\text{ECEF}} = B_{\text{ECEF}}/\mu_0$, where μ_0 is the magnetic permeability of free space. H_{ECEF} is rotated to the body frame via $H_{\text{BI}} = R_{\text{BI}} R_{\text{EI}}^T H_{\text{ECEF}}$, which supplies the hysteresis model. It provides the body-frame flux density $B = H_{\text{BI}} \mu_0$ for the permanent magnet torque calculations: $\tau = m \times B$, wherein m is the magnetic dipole. Environmental torques such as atmospheric drag, solar radiation pressure, and gravity gradient are excluded in the present model, and it is noted that an implementation with environmental torques is necessary for making steady-state predictions for LEO. At each step of the integration loop, the passive magnetic torques are evaluated and applied to advance the system state. The first momentum kick is applied, calculating the next half-momentum by the following, see equation (14):

$$\Pi_{k+\frac{1}{2}} = \Pi_k + \frac{h}{2} M_k. \quad (17)$$

Third, a Newton iteration method is utilized to find the rotation increment vector f_k required for the attitude update. Before the loop begins, an initial guess for f_k using $f_{k,0} = h(J^{-1}\Pi_{k+\frac{1}{2}})$ is made. Within the loop, the discrete residual G is calculated using (16). A Jacobian matrix DG is constructed by differentiating G with respect to f_k . The rotation increment is updated within the solver by the equation

$$f_{k,i+1} = f_{k,i} - (DG)^{-1}G. \quad (18)$$

This process is repeated until the magnitude of the residual reaches a specified tolerance, for example $tol = 1 \times 10^{-12}$ such that $\|G(f_k)\| < tol$. Then, (15) maps f_k to F_k , and the attitude update (2) is applied in the drift sequence. An analysis of the Newton iteration implemented in the simulator found that the algorithm performed exactly two iterations before f_k converged for every call. Fourth, the future external torques are calculated by the same process described in step two, except for simulation time t_{k+1} . The future angular momentum is calculated via the second kick using equation (14):

$$\Pi_{k+1} = F_k^T \Pi_{k+\frac{1}{2}} + \frac{h}{2} M_{k+1}. \quad (19)$$

Fifth, diagnostic data is logged for debugging. It is recommended to log variables angular velocity, pointing error, angular momentum, and orthogonality error, which are used to determine the time to steady-state attitude and to verify the physical integrity of the simulator. The integration loop continues for the duration of the simulation, propagating the satellite's attitude according to Hamilton's principle. In the absence of external torques, momentum is inherently preserved. However, the introduction of magnetic torques from the PMAC system components causes the body's angular momentum to change over time. Through the mechanism of hysteresis damping, the system's rotational energy dissipates as it is converted to heat.

MAGNETIC HYSTERESIS

A CubeSat is deployed from the International Space Station with initial rotational energy that a PMAC system removes to reach and maintain steady-state attitude. Magnetic hysteresis is a process that accounts for the dissipation of rotational energy supplied by hysteresis rods, a primary component of PMAC systems. As discussed by Gerhardt in his work on an RK45 attitude damping simulation for the Colorado Student Space Weather Experiment (CSSWE), the magnetic torques that drive hysteresis damping are produced by shifting polarities within ferromagnetic rods.³ These shifts occur in delayed response to magnetic field changes, resulting in the conversion of rotational energy to heat.

A hysteresis loop represents the relationship between the magnetic field strength H applied to a hysteresis rod and the magnetic flux density B induced within the rod. The area of the loop determines the energy dissipation per unit volume, per cycle. It is shaped by three parameters of the rods: the coercive force H_c , the remanence B_r , and the saturation induction B_s . These parameters represent different characteristics of the hysteresis loop; saturation induction determines the maximum height of the loop, and the coercive force and remanence define the X- and Y-intercepts. Generally, higher H_c , B_r , and B_s values correlate with larger loop area, resulting in increased induced flux density and damping rates. This nonlinear behavior is modeled at each simulation timestep.

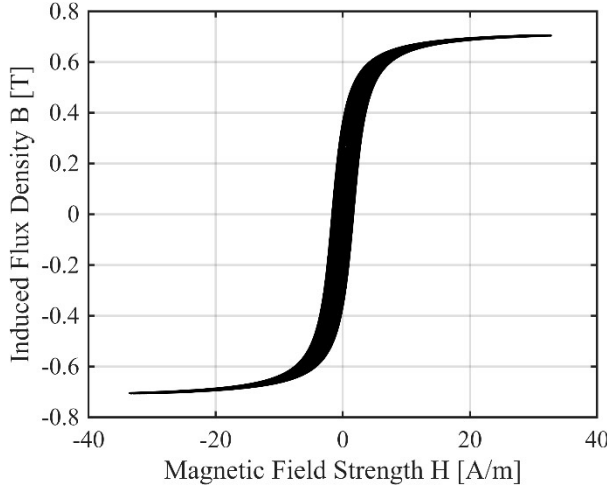


Figure 3. Reproduced Hysteresis Loop using Park et al.'s Simulation Parameters

Hysteresis models output the induced flux density B , which is used to find the rod's magnetic moment m . The torque τ exerted by the rod is found via $\tau = m \times B$. A model developed by Flatley and Henretty is widely adopted, which closely approximates empirically determined hysteresis data.⁴ The model being nonlinear and multivalued, it is expressed in its piecewise form as

$$B = \frac{2}{\pi} B_s \tan^{-1} \left[\frac{1}{H_c} \tan(k_1)(H \pm H_c) \right], \quad (20)$$

where the constant k varies by implementation:

$$k_1 = \left(\frac{\pi B_r}{2B_s} \right), \quad k_2 = \left(\frac{\pi B}{2B_s} \right).$$

Used by Gerhardt in his simulations for CSSWE, this model was formulated by conjoining two arctangent curves fitted to hysteresis parameters, producing a multivalued result. The $\pm$ operator depends on the time derivative of the applied field strength, $\dot{H}$. A limitation of this algebraic model is the discontinuities at the vertices of the loop that introduce artificial energy changes.

A more accurate representation is the differential version used by Park et al. in their LGVI attitude damping simulations for RAX.⁵ In this method, B becomes a state variable integrated by the solver, and the upper and lower branches are drawn from *sin* and *cosine* functions:

$$\dot{B} = \frac{2B_s}{H_r \pi} \left(\frac{(H \pm H_c) \cos(k_2) \mp H_r \sin(k_2)}{2H_c} \right)^2 \dot{H}. \quad (21)$$

(21) provides a continuous model compatible with the fixed-step LGVI and is implemented via a first-order forward Euler scheme. The Euler update outputs B for the next timestep and can be expressed as

$$B_{k+1} = B_k + h \cdot \dot{B}(H_k, \dot{H}_k, B_k). \quad (22)$$

This differential model treats the magnetic induction of the rod as a dynamic state and provides a more physically accurate energy dissipation effect.

SIMULATOR VALIDATION

Two attitude simulators are developed in MATLAB for the purpose of evaluating the performance of proposed PMAC system configurations and informing the selection of the dimensional sizing for GAS-RATS. The simulators are based on the architectures of the CSSWE and RAX PMAC simulations. The development follows a three-step validation chain. This process is necessary for ensuring physical fidelity, and it resulted in two validated simulators coupling numerical attitude integrators with hysteresis and geomagnetic models. The baseline RK45 simulator incorporates MATLAB's ode45 solver with a piecewise, algebraic Flatley-Henretty hysteresis model, based on the architecture discussed by Gerhardt. It utilizes a stable quaternion formulation that avoids trigonometric singularities present in Gerhardt's Euler-angle kinematic equations. The second simulator incorporates a second-order LGVI with a differential Flatley-Henretty hysteresis model, following the architecture proposed by Park et al.

Reference Validation

The developed RK45 and LGVI attitude simulators are validated against the literature to test the fidelity of the encoded physics by comparing convergence behavior. Upon initializing the same CSSWE parameters, the angular velocity damping profile of the RK45 implementation reproduces key features of Gerhardt's results. A shoulder region of the reference angular velocity reflects similar structure and location. The expected consolidation of angular

velocity components into individual strands and subsequent divergence and reconvergence is evident. Additionally, the time to steady-state attitude is in close agreement with the literature. For the LGVI simulator configured on the same RAX parameters, the damping profile and evolving shape of the pointing error align with Park et al.'s results, see Fig. 4. The settling times are comparable, with a variance of one simulated orbit that is attributed to unspecified initial orbital geometry in the reference study. The validation confirms a satisfactory alignment between the present simulators and references, qualifying for integration with GASRATS parameters.

Cross-Framework Verification

Having been validated against their respective source literature, a cross-framework verification study is conducted to assess the consistency of damping behavior between simulators. Identical convergence patterns are not expected because the architectures are different. However, this study is valuable for discovering sensitivities associated with each implementation. Simulation parameters for GASRATS are used (see Table 1). The hysteresis parameters for GASRATS are determined by comparison to Gerhardt's hysteresis parameters. The same apparent coercive force and apparent remanence are adopted, as GASRATS uses the same rod alloy, HyMu-80. B'_s was calculated via a procedure provided by Gerhardt that is beyond the scope of this paper. Apparent hysteresis parameters are implemented because they account for the demagnetizing effect of the rod's own magnetic field. While this verification step employs apparent properties, the previous LGVI validation with Park et al.'s Scenario 4 uses intrinsic properties. Therefore, the resulting angular velocity profile in Fig. 4 represents accelerated damping from idealized rods (see also Fig. 3).

The RK45 and LGVI simulators are configured to a maximum timestep of 0.05 seconds. Additionally, the LGVI simulator implements the 14th generation of the International Geomagnetic Reference Field (IGRF-14) model following Park et al.'s architecture. The field is precomputed at a coarse interval of one second to save computation time. The simulators are evaluated based on the consistency of the

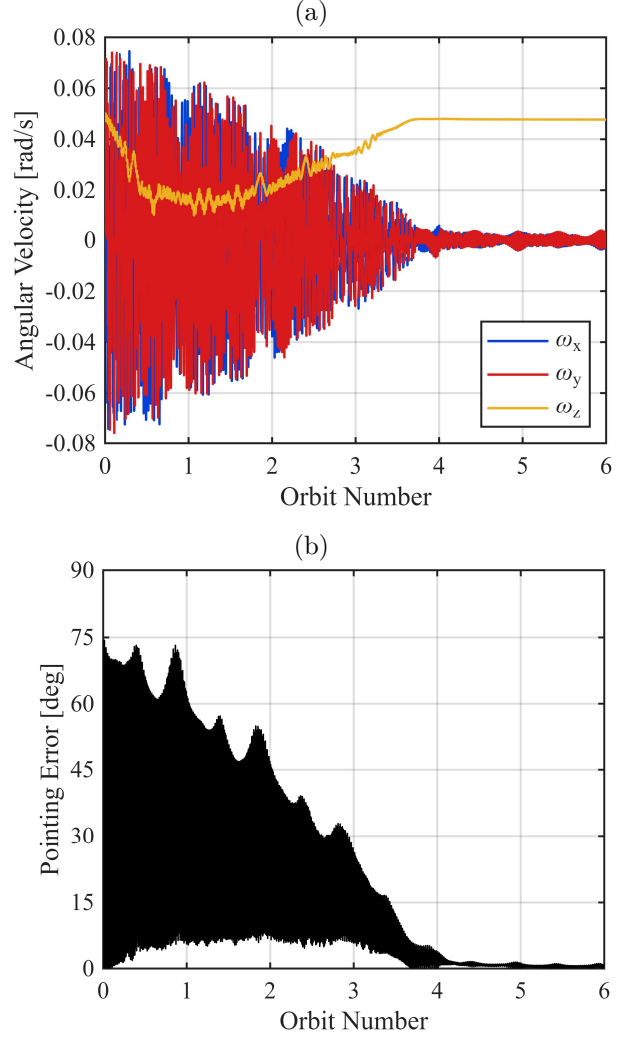


Figure 4. Reproduced LGVI Simulation Results of Park et al.'s Scenario 4: (a) Angular Velocity, (b) Pointing Error

angular velocity and pointing error damping profiles and on the time to steady-state attitude (see Fig. 5). The steady-state criteria for GASRATS are driven by the mission payload, a transparent patch antenna. Patch antennas possess narrower beamwidth than omnidirectional antennas, and a more stringent settling requirement may be needed to ensure link closure during ground station passes. Although a downlink analysis is beyond the scope of this paper, three candidate settling criteria are considered. While Gerhardt's simulations were evaluated using a steady-state criterion of maintaining pointing error $\beta < 15^\circ$, the steady-state requirements for this study are defined as maintaining $\|\omega\| < 0.5 \text{ deg/s}$ and $\beta < 10^\circ$.

Table 1. Initial GASRATS Parameters for Cross-Framework Verification

Parameter	Symbol	Value
Altitude [km]	h	400
Inclination [deg]	i	51.6
Initial Angular Velocity [deg/s]	ω_0	[10, 5, 5]
Initial Orientation	q_0	[1, 0, 0, 0]
Mass [kg]	m	2
Transverse Inertia [$\text{kg} \cdot \text{m}^2$]	J_{xx}, J_{yy}	0.0093
Axial Inertia [$\text{kg} \cdot \text{m}^2$]	J_{zz}	0.0033
Permanent Magnet Strength [$\text{A} \cdot \text{m}^2$]	m_{pm}	0.3, 0.03
Number of Rods	n	4
Total Rod Volume [m^3]	V	5.58×10^{-7}
Apparent Coercive Force [A/m]	H'_c	12
Apparent Remanence [T]	B'_r	0.004
Apparent Saturation Induction [T]	B'_s	0.0128

The resulting profiles of the angular velocity and pointing error displayed in Fig. 5 demonstrate attitude convergence but do not exhibit rigorous qualitative agreement. This is expected and can be attributed to the architectural differences between simulators; different magnetic field and hysteresis models are used. Additionally, the LGVI simulator is configured with candidate magnet strength $0.03 \text{ A} \cdot \text{m}^2$ as the $0.3 \text{ A} \cdot \text{m}^2$ configuration exhibited oscillatory nonconvergent behavior for Table 1 parameters. The RK45 simulation reaches steady-state attitude in 7.29 days, while the LGVI simulation does so in 12.42 days. This delayed convergence is caused by the bounded oscillatory pattern of the LGVI simulation's pointing error after day three that marginally exceeds the 10° threshold. Thus, it is determined that this pointing error threshold is overly stringent, and a 15° requirement consistent with Gerhardt's methodology is adopted going forward.

Invariant-Preservation Analysis

The final validation step is an invariant-preservation analysis that tests the conservation of kinetic energy and angular momentum and the preservation of the norm and orthogonal structure of quaternions and attitude matrices in the LGVI and RK45. The test configuration adopts a torque-free adaptation of the simulators used in the cross-framework validation. A simulation duration of 50 orbits or 3.21 days is implemented.

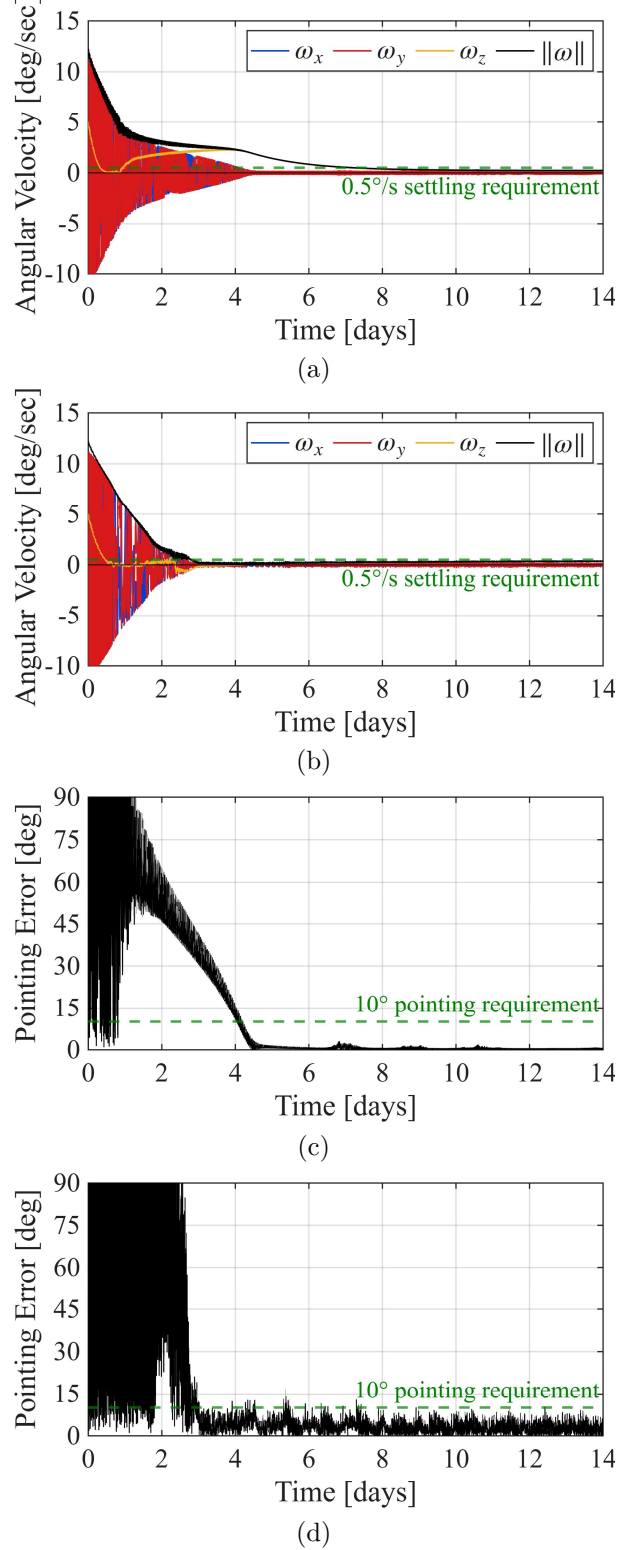


Figure 5. Cross-Framework Verification: (a) RK45 Angular Velocity, (b) LGVI Angular Velocity, (c) RK45 Pointing Error, (d) LGVI Pointing Error

The test reveals that both the RK45 and LGVI frameworks achieve a high degree of fidelity in invariant preservation, and the RK45 integrator outperforms initial expectations. This is attributed to the RK45's restricted maximum timestep of 0.05 seconds that corresponds with the LGVI's fixed timestep. Both models exhibit desirable structure preservation, approaching relative orthogonality and norm errors on the order of 10^{-12} . While the LGVI demonstrates superior angular momentum conservation with a relative error on the order of

10^{-13} , the RK45 approaches a relative momentum error on the order of 10^{-9} . This figure is expected to decline further for longer simulation timespans. Although the LGVI shows lower kinetic energy preservation for the tested duration, it more accurately preserves angular momentum to a significant degree while maintaining rotation-matrix orthogonality of 10^{-11} relative error. Therefore, the LGVI is selected for use in the Monte Carlo analysis, with kinetic energy error retained as an implementation limitation. Additionally, the torque-free LGVI performs the 50-orbit test in 51.3 seconds, halving the runtime of the RK45.

MONTE CARLO ANALYSIS

A 1,001-trial Monte Carlo analysis is conducted on GASRATS mission parameters to inform key mission design decisions such as the hysteresis rod count and satellite size. This comprehensive test utilizes the validated LGVI simulator discussed in previous sections, which is selected for its angular momentum preservation (see Fig. 6) and computational efficiency. As preliminary testing indicates the inadequate hysteresis damping provided by two rods, the computational power available for the test is focused on configurations with four rods (801 trials versus 200).

The trial sets are split quarterly towards each CubeSat configuration (1U, 1.5U, 2U, 3U). The additional four-rod trial is a control trial using the same parameters detailed in the cross-framework verification study (see Table 1) for verifying batch integrity. MATLAB's Parallel Computing Toolbox was used to distribute the Monte Carlo trials across eight workers on a 13th Gen Intel® Core™ i7-13700K CPU. Initial angular velocity magnitudes range from zero to $35^\circ/\text{s}$, whose upper bound represents the nominal deployment velocity limit $\|(5^\circ/\text{s}, 5^\circ/\text{s}, 5^\circ/\text{s})\|$ from the Nanoracks CubeSat Deployer multiplied by a safety factor of four.

Preliminary testing reveals that an angular velocity asymptote at $\|\omega\| = 0.5^\circ/\text{s}$ exists for certain cases, causing false non-convergence. To ensure accurate convergence reporting, the settling requirement of $\|\omega\| < 0.5^\circ/\text{s}$ is raised to $\|\omega\| < 0.75^\circ/\text{s}$. Each trial is run for 30 simulated days, and a trial is categorized

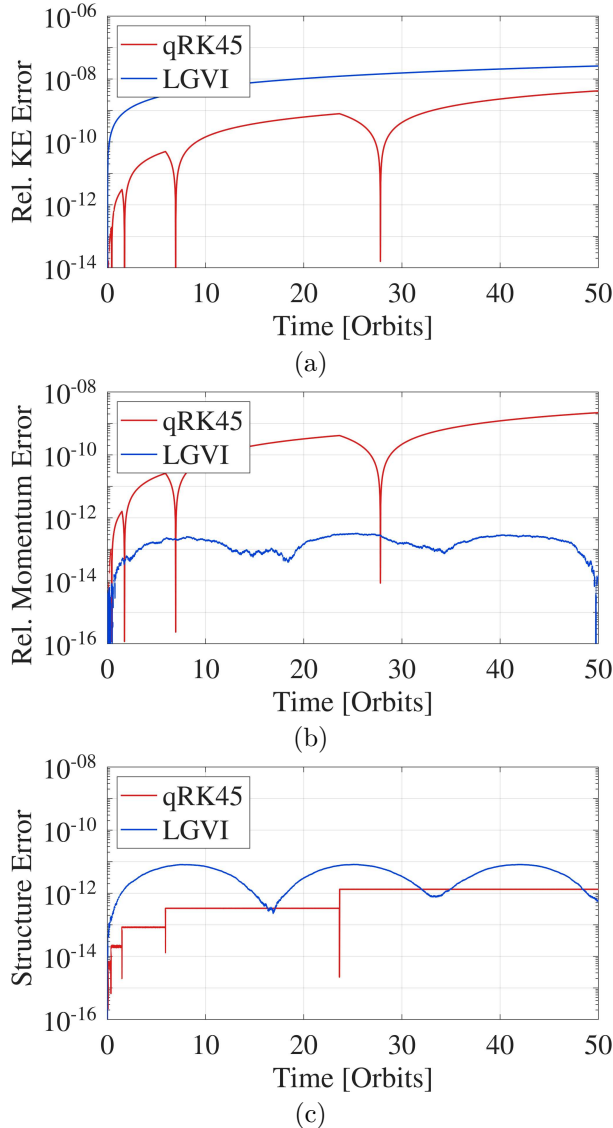


Figure 6. Invariant-Preservation Analysis Results: (a) Relative Kinetic Energy Error, (b) Relative Momentum Error, (c) Orthogonality/Norm Error

as successful if the steady-state criteria of $\|\omega\| < 0.75^\circ/s$ and $\beta < 15^\circ$ is reached within the mission requirement of seven days and maintained until the end of the simulation.

CONCLUSIONS

The results of the analysis (see Fig. 7 and Table 2) confirm expectations for configurations with two rods to produce insufficient damping. Across the initial angular velocity range of 0 to $35^\circ/s$, 50% of trials with two rods reach steady-state attitude within the mission requirement, while 86% of trials with four rods reach steady-state within the mission requirement. The 1.5U configuration demonstrates a 100% success rate for settling within mission requirements for initial angular velocities up to three times the nominal velocity limit. These results inform the final recommendation of a four-rod PMAC system and 1.5U size selection for GASRATS.

Based on the results, it is determined that higher initial rotational kinetic energy corresponds with longer damping times. As initial rotational kinetic energy scales with velocity and inertia, it is expected for the settling duration to increase as initial angular velocity and inertia increase. This trend is realized for sizes 1.5U, 2U, and 3U, whereas the 1U case is an anomaly. Although the 1U was expected

Table 2. 4-Rod Monte Carlo Results

Configuration / Bin	Pass Rate	Percentage
GASRATS 1U / 4-Rod Configuration		
Bin 1: $00.00^\circ/s \leq \ \omega\ < 08.66^\circ/s$	51/56	91.07%
Bin 2: $08.66^\circ/s \leq \ \omega\ < 17.32^\circ/s$	30/39	76.92%
Bin 3: $17.32^\circ/s \leq \ \omega\ < 25.98^\circ/s$	29/54	53.70%
Bin 4: $25.98^\circ/s \leq \ \omega\ < 35.00^\circ/s$	25/51	49.02%
GASRATS 1.5U / 4-Rod Configuration		
Bin 1: $00.00^\circ/s \leq \ \omega\ < 08.66^\circ/s$	49/49	100.00%
Bin 2: $08.66^\circ/s \leq \ \omega\ < 17.32^\circ/s$	51/51	100.00%
Bin 3: $17.32^\circ/s \leq \ \omega\ < 25.98^\circ/s$	46/46	100.00%
Bin 4: $25.98^\circ/s \leq \ \omega\ < 35.00^\circ/s$	48/54	88.89%
GASRATS 2U / 4-Rod Configuration		
Bin 1: $00.00^\circ/s \leq \ \omega\ < 08.66^\circ/s$	53/53	100.00%
Bin 2: $08.66^\circ/s \leq \ \omega\ < 17.32^\circ/s$	51/51	100.00%
Bin 3: $17.32^\circ/s \leq \ \omega\ < 25.98^\circ/s$	44/47	93.62%
Bin 4: $25.98^\circ/s \leq \ \omega\ < 35.00^\circ/s$	37/49	75.51%
GASRATS 3U / 4-Rod Configuration		
Bin 1: $00.00^\circ/s \leq \ \omega\ < 08.66^\circ/s$	54/54	100.00%
Bin 2: $08.66^\circ/s \leq \ \omega\ < 17.32^\circ/s$	51/51	100.00%
Bin 3: $17.32^\circ/s \leq \ \omega\ < 25.98^\circ/s$	37/44	84.09%
Bin 4: $25.98^\circ/s \leq \ \omega\ < 35.00^\circ/s$	34/52	65.38%

to consistently reach steady-state within seven days due to its low inertia, this configuration demonstrates the poorest damping success rate out of the four-rod trials. This phenomenon can be attributed to the 1U's inertial symmetry.

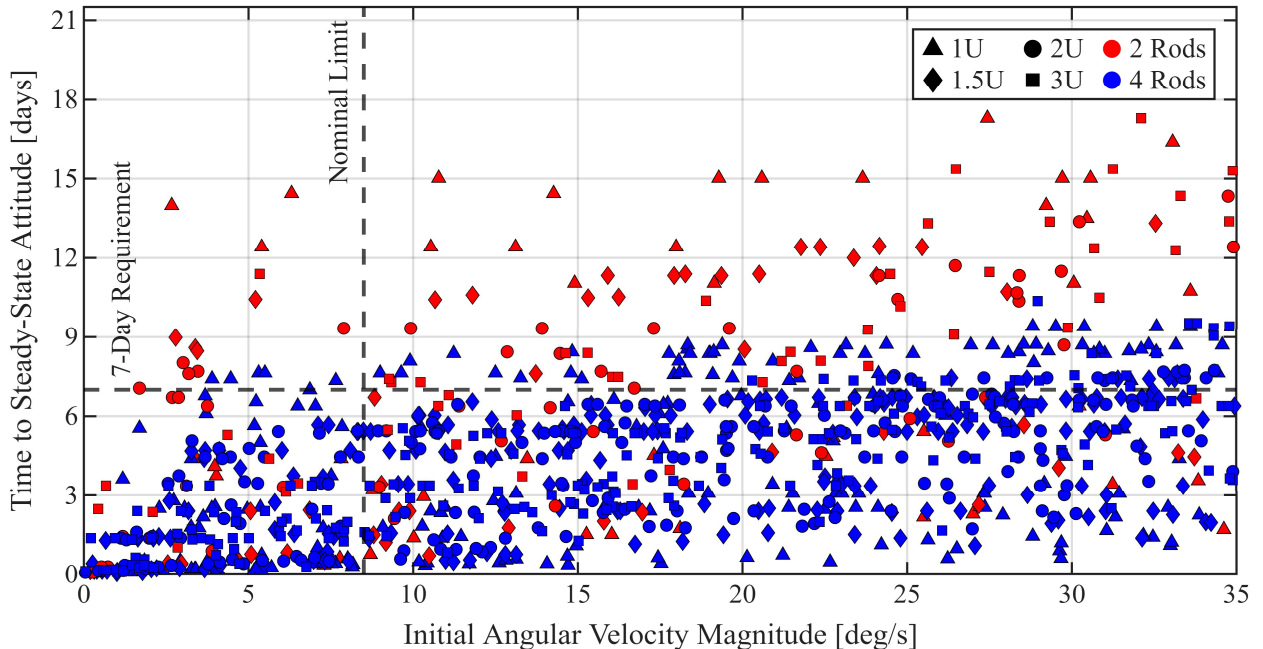


Figure 7. 1,001-Trial Monte Carlo Analysis for GASRATS

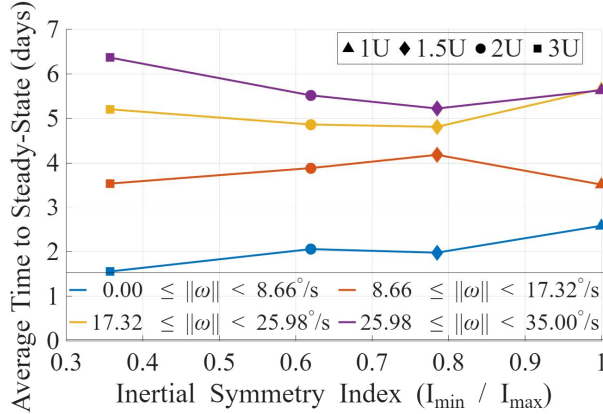


Figure 8. Inertial Symmetry vs. Settling Time for 4-Rod Trials

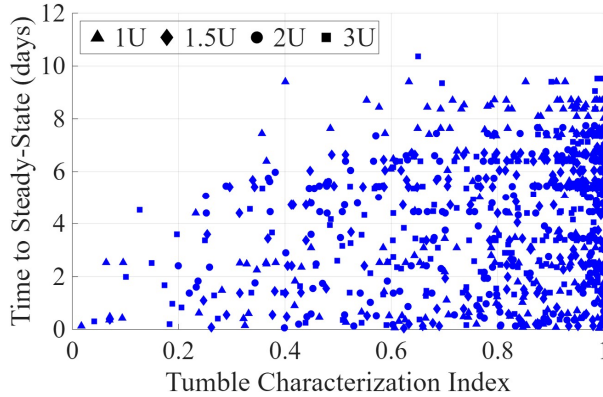


Figure 9. Tumble Characterization Index vs. Settling Time for 4-Rod Trials

Index 0 indicates a pure Z-axis spin, and index 1 indicates a pure X-Y tumble.

A rotating body with inertial symmetry experiences limited rotational energy transfer between body axes. Hence, 1U configurations respond poorly to non-ideal deployment scenarios such as end-over-end tumbles wherein angular velocity components on the X- and Y-axes dominate the Z-component velocity. As an arrangement of hysteresis rods on the X- and Y-axes primarily removes rotational energy from the Z-axis, an end-over-end tumble for a CubeSat with inertial symmetry realizes less efficient damping. Fig. 8 and Fig. 9 represent the effect of inertial symmetry and tumble type on settling time. Notice in Fig. 8 that for average nominal deployment velocities, the 1U configuration demonstrates the longest average damping time despite having the least inertia. Conclusions from the Monte Carlo study are limited to comparative

analysis between configurations and do not provide reliable estimates for damping times in LEO.

FUTURE WORK

Dynamic environmental torques will be implemented to provide defensible estimates for settling times in LEO. Beyond Euclidean spaces, LGVIs can be formulated on relativistic manifolds, such as Minkowski spacetime and the coadjoint orbit of the Poincaré group, for simulators that preserve relativistic effects such as time dilation and Thomas precession.

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Code Availability:

github.com/TrumanTAbbe/CubeSat-LGVI-Magnetic-Attitude-Control-Simulators

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